

MEMBER 1 — PROBLEM

“Good morning respected judges and mentors.

We are **Team VS TECH**, and our solution is **EchoVersity**, for problem statement **SIH26104 — Real-Time AI Detection and Prevention of Voice-Cloning Impersonation Attacks**.

Let me start with a simple situation.

Imagine you receive a call from your parent or a close friend. The voice sounds exactly like them, and they urgently ask you to send money.

Would you trust that voice?

Today, AI can generate voices that sound very similar to real people. Attackers can use these voices for impersonation, scams, and financial fraud.

The biggest problem is that people usually realize the voice was fake **only after the damage is done**.

So our problem is not simply creating a fake voice.

The real problem is: how do we identify that fake voice while the conversation is happening?

That is the gap we are addressing with EchoVersity.

Now, my teammate will explain the existing approach and what we are proposing differently.”

MEMBER 2 — EXISTING APPROACH + SOLUTION

“Thank you.

Existing anti-spoofing research already gives us strong models and datasets for detecting synthetic speech. But many detection approaches are mainly used for offline analysis or controlled testing.

Our focus is different.

We want to bring detection closer to the real-time conversation.

Our solution, EchoVersity, works like a spam filter for your ears.

It checks small parts of the conversation and looks for patterns that may indicate an AI-generated voice.

Our process has four steps:

Detect — analyse the voice for synthetic patterns.

Alert — give the user a warning with a confidence score.

Verify — if the risk is high, ask the speaker to repeat a random phrase.

Protect — store verified fraud patterns in a shared registry that can help organisations improve their defence.

Technically, we are proposing an edge-first approach, so the main detection can happen on the device, reducing dependence on the cloud and helping with privacy.

Our target is fast detection using a lightweight, quantized model.

Now, my teammate will explain why this idea is practical and the impact it can create.”

MEMBER 3 — FEASIBILITY + IMPACT + CONCLUSION

“Thank you.

We believe EchoVersity is practical to develop as an MVP.

We can start with public datasets such as ASVspoof, WaveFake and In-The-Wild, and use established anti-spoofing architectures such as AASIST and RawNet2.

We also understand the limitations.

AI-generated voices keep improving, phone-call compression can affect detection, and false positives are possible.

So we plan regular model updates, testing with compressed audio, and confidence-based alerts instead of automatically blocking genuine calls.

For privacy, our approach is on-device first, so raw audio does not need to leave the device by default.

The impact can extend across families, banks, fintech companies, telecom and law enforcement.

But our goal is not to claim that we can detect every fake voice.

Our goal is to build a practical additional layer of protection that can improve as the threat evolves.

So we want to end with one simple thought:

Today, hearing someone's voice is no longer enough to prove who they are.

That is why our vision is:

“Don't trust the voice. Verify the voice.”

We are Team VS TECH, and this is EchoVersity.

Thank you, respected judges. We are ready for your questions.”

1. What exactly is EchoVersity?

“EchoVersity is a real-time voice-cloning detection system. It analyses short parts of a conversation, identifies patterns that may indicate synthetic speech, and gives the user a warning with a confidence score.”

2. What is unique about your solution?

“Our uniqueness is not simply using AI for voice detection. We combine real-time detection, edge-based processing, confidence-based alerts, optional liveness verification, and shared fraud intelligence in one workflow.”

3. How does your AI identify a cloned voice?

“The system analyses characteristics of the audio such as pitch, spectral patterns and other acoustic features. The anti-spoofing model then estimates whether the speech is more likely to be genuine or synthetic.”

4. Which AI model are you using?

“For our implementation, we are considering established anti-spoofing architectures such as AASIST and RawNet2. We will select and fine-tune the most suitable model based on our validation results.”

5. What datasets will you use?

“We plan to use ASVspoof , WaveFake, and the In-The-Wild deepfake audio dataset.”

6. What happens if the attacker creates a new type of AI voice?

“That is one of our main challenges. We cannot assume that one model will detect every future voice-generation method. So our plan is to continuously update and retrain the detector using new samples and new attack types.”

7. Can you really detect a cloned voice in real time?

“Our architecture is designed for real-time detection using short rolling audio windows and lightweight edge inference. Our target is below 150 milliseconds for the quantized inference stage. We still need to validate the actual end-to-end performance during implementation.”

8. How will you access live phone-call audio?

“We access the live audio through the call application's or telecom platform's audio SDK/API. It captures the audio stream, and EchoVersity analyses it in small rolling windows in real time.”

9. Isn't continuously analysing calls a privacy problem?

“Yes, privacy is an important concern. That's why our design is on-device first. Our goal is that raw audio should not leave the device by default. We only use additional processing when necessary and permitted.”

10.What will be the cost of your project?

The major cost will range from 3-5 thousand.

11.why did you use ASVspoof,WaveFake and in the world?

These are all the existing dataset gives us a large collection of generated audio from different speech-generation systems, so it helps us learn different synthetic voice patterns.

12.why AASIST and RawNet2.

Because both are specifically designed for audio anti-spoofing, which matches our problem. **RawNet2** works directly with the raw audio waveform and can learn useful spoofing patterns **AASIST** goes one step further by modelling both **time and frequency-related patterns** in the speech

13.What is anti-spoofing?

Anti-spoofing means detecting whether the voice we are hearing is genuine or artificially generated. In our project, it helps us identify cloned or synthetic voices before the user trusts the call.